Discrete Structures: The Foundations – Logic and Proofs

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Outline

- Propositional Logic
- Applications of Propositional Logic
- Propositional Equivalences
- Predicates and Quantifiers
- Nested Quantifiers
- Rules of Inference
- Introduction to Proofs

A Sample Question

- When three professors are seated in a restaurant, the hostess asks them: "Does everyone want coffee?"
- The first professor says: "I do not know."
- The second professor then says: "I do not know."
- Finally, the third professor says: "No, not everyone wants coffee."
- The hostess comes back and gives coffee to the professors who want it.

Basic Inference Form 1

- Example
  - "If you have the current password, then you can log in."
    "You have the current password."
    Therefore,
    "You can log in."
- We can imagine the typical inference looks like:

$$(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow q$$

  - A sequence of facts can be used to infer further conclusion
  - Facts: pre-known true statements/propositions/**premises**
    - Each proposition/fact p_i could also be any compound proposition
  - **Conclusion**: inferred proposition

Basic Inference Form 2

- Wrong example
  - A happened before B, therefor A caused B.
  - This is an invalid argument, and wrong inference (fallacy).
- Formal definition:

Term	Definition
argument	A sequence of statements that ends with a conclusion.
valid	The conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or premises, of the argument.
fallacy	Incorrect reasoning

Rules of Inference 1

- A rule of inference is also in the form of

$$(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow q$$

- To make it a rule, this entire proposition needs to be a tautology
  - A tautology is always true
- Recall the previous example

English Statements	Math Logics
"If you have the current password, then you can log in."	$p \rightarrow l$
"You have the current password."	p
Therefore, "You can log in."	$\therefore l$

  - The logical representation of this argument $((p \rightarrow l) \wedge p) \rightarrow l$
  - Is this following a rule of inference? Let's get its truth table.

Rules of Inference 2

- Truth table for $((p \rightarrow l) \wedge p) \rightarrow l$

p	l	$p \rightarrow l$	$(p \rightarrow l) \wedge p$	$((p \rightarrow l) \wedge p) \rightarrow l$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

- So $((p \rightarrow l) \wedge p) \rightarrow l$ is a tautology, and can be used as a rule for inference
  - This tautology is very intuitive.
  - Following this way, the conclusion is reliable.

Rules of Inference 3

Rule of Inference		Equivalent Tautology	Name
	p	$[\boldsymbol{p} \wedge (\boldsymbol{p} \rightarrow \boldsymbol{q})] \rightarrow \boldsymbol{q}$	**Modus ponens**
	$p \rightarrow q$		
$\therefore$	q		
	$\neg q$	$[\neg \boldsymbol{q} \wedge (\boldsymbol{p} \rightarrow \boldsymbol{q})] \rightarrow \neg \boldsymbol{p}$	**Modus tollens**
	$p \rightarrow q$		
$\therefore$	$\neg p$		
	$p \rightarrow q$	$[(\boldsymbol{p} \rightarrow \boldsymbol{q}) \wedge (\boldsymbol{q} \rightarrow \boldsymbol{r})] \rightarrow (\boldsymbol{p} \rightarrow \boldsymbol{r})$	Hypothetical syllogism
	$q \rightarrow r$		
$\therefore$	$p \rightarrow r$		
	$p \vee q$	$[(\boldsymbol{p} \vee \boldsymbol{q}) \wedge \neg \boldsymbol{p}] \rightarrow \boldsymbol{q}$	Disjunctive syllogism
	$\neg p$		
$\therefore$	q		

Rules of Inference 4

Rule of Inference		Tautology	Name
	p	$\boldsymbol{p} \rightarrow (\boldsymbol{p} \vee \boldsymbol{q})$	Addition
$\therefore$	$p \vee q$		
	$p \wedge q$	$(\boldsymbol{p} \wedge \boldsymbol{q}) \rightarrow \boldsymbol{p}$	Simplification
$\therefore$	p		
	p	$[(\boldsymbol{p}) \wedge (\boldsymbol{q})] \rightarrow (\boldsymbol{p} \wedge \boldsymbol{q})$	Conjunction
	q		
$\therefore$	$p \wedge q$		
	$p \vee q$	$[(\boldsymbol{p} \vee \boldsymbol{q}) \wedge (\neg \boldsymbol{p} \vee \boldsymbol{r})] \rightarrow (\boldsymbol{q} \vee \boldsymbol{r})$	**Resolution**
	$\neg p \vee r$		
$\therefore$	$q \vee r$		

Rules of Inference 5

Rule of Inference		Name
	$\forall x P(x)$	Universal instantiation
$\therefore$	$P(C)$	
	$P(C)$ for an arbitrary C	Universal generalization
$\therefore$	$\forall x P(x)$	
	$\exists x P(x)$	Existential instantiation
$\therefore$	$P(C)$ for some element C	
	$P(C)$ for some element C	Existential generalization
$\therefore$	$\exists x P(x)$	

Modus Ponens

$$\begin{array}{ll} & p \\ & p \to q \\ \hline \therefore & q \end{array} \qquad (p \wedge (p \to q)) \to q$$

- **Example**: Given the propositions: "If it snows today, then we will go skiing" and "It is snowing today", are we going to skiing?

	q	We will go skiing
	p	It is snowing today.
	$p \to q$	If it snows today, then we will go skiing.
$\therefore$	q	Therefore, we will go skiing

Modus Tollens 1

$$\begin{array}{ll} & \neg q \\ & p \to q \\ \hline \therefore & \neg p \end{array}$$

$$[\neg \boldsymbol{q} \wedge (\boldsymbol{p} \to \boldsymbol{q})] \to \neg \boldsymbol{p}$$

- **Example**: Given the propositions: "If it snows today, the university will be closed" and "The university will not be closed today", then does it snow today?

	p	It snows today.
	q	The university will be closed.
	$\neg q$	The university will not be closed today.
	$p \to q$	If it snows today, then university will be closed.
$\therefore$	$\neg p$	Therefore, it doesn't snow today.

Modus Tollens 2

- Modus Tollens is connected with Modus Ponens by **equivalent contrapositive**
  - We know this commonly used equivalence about contrapositive:
    $\boldsymbol{p \rightarrow q \equiv \neg q \rightarrow \neg p}$

Modus Tollens	Contrapositive	Substitute	Modus Ponens
$\neg q$	$\boldsymbol{\neg q}$	s	p
$p \rightarrow q$	$\boldsymbol{\neg q \rightarrow \neg p}$	$s \rightarrow t$	$p \rightarrow q$
$\therefore \neg p$	$\boldsymbol{\therefore \neg p}$	$\therefore t$	$\therefore q$

Resolution 1

$$\begin{array}{ll} & p \vee q \\ & \neg p \vee r \\ \hline \therefore & q \vee r \end{array}$$

$$[(\boldsymbol{p} \vee \boldsymbol{q}) \wedge (\neg \boldsymbol{p} \vee \boldsymbol{r})] \rightarrow (\boldsymbol{q} \vee \boldsymbol{r})$$

- **Example**: Given the propositions: "It is not snowing or Jasmine is skiing", and "It is snowing or Bart is playing hockey", then can you tell what are Jasmine and Bart doing?

	p	It is snowing.
	q	Jasmine is skiing.
	r	Bart is playing hockey.
	$\neg p \vee q$	It is not snowing or Jasmine is skiing.
	$p \vee r$	It is snowing or Bart is playing hockey.
$\therefore$	$q \vee r$	Therefore, Jasmine is skiing or Bart is playing hockey.

Resolution 2

- This is a discussion/resolution of p's values
  - If p is true, then q happens, otherwise r happens
  - No matter what's the value of p, then q or r would happen.
- Recall another common equivalence

$$\boldsymbol{p \rightarrow q \equiv \neg p \vee q}$$

Resolution	Equivalence
$\neg p \vee q$	$p \rightarrow q$
$p \vee r$	$\neg p \rightarrow r$
$\therefore q \vee r$	$\therefore q \vee r$

Examples of Inference 1

- **Example**: Show that the hypotheses "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be at home by sunset."
- **Solution**:

p	It is sunny this afternoon.
q	It is colder than yesterday.
r	We will go swimming.
s	We will take a canoe trip.
t	We will be home by sunset.

$$\neg p \wedge q$$
$$r \rightarrow p$$
$$\neg r \rightarrow s$$
$$s \rightarrow t$$

Examples of Inference 2

Step	Proposition	Explanation
1	$\neg p \wedge q$	**Hypothesis**
2	$\neg p$	Simplification (of step 1)
3	$r \rightarrow p$	**Hypothesis**
4	$\neg r$	Modus tollens using steps 2 and 3 $[\neg \boldsymbol{p} \wedge (\boldsymbol{r} \rightarrow \boldsymbol{p})] \rightarrow \neg \boldsymbol{r}$
5	$\neg r \rightarrow s$	**Hypothesis**
6	s	Modus ponens using steps 4 and 5 $[\neg \boldsymbol{r} \wedge (\neg \boldsymbol{r} \rightarrow \boldsymbol{s})] \rightarrow \boldsymbol{s}$
7	$s \rightarrow t$	**Hypothesis**
8	t	Modus ponens using steps 6 and 7.$[\boldsymbol{s} \wedge (\boldsymbol{s} \rightarrow \boldsymbol{t})] \rightarrow \boldsymbol{t}$

- **How can we know this order: think about the reversed procedure**

Examples of Inference 3

- Think in a **reversed** way: help us trace back

Step	Proposition	Explanation
1	$\neg p \wedge q$	**Hypothesis**
2	$r \rightarrow p$	**Hypothesis**
3	$\neg r \rightarrow s$	**Hypothesis**
4	$s \rightarrow t$	**Hypothesis**
5	t	**Conclusion**. The only appearance of t in the premises is in line 4. To have t, we need s
6	s	Other than line 4, the only appearance of s is in line 3, and to have s , we need $\neg r$
7	$\neg r$	Other than line 3 and line 4, the only appearance $\neg r$ of is in line 2, and to have $\neg r$, we need $\neg p$
8	$\neg p$	$\neg p$ is part of line 1

Examples of Inference 4

- **Fallacies**: Incorrect reasoning

Proposition	Description
$[(p \to q) \land q] \to p$	Fallacy of affirming the conclusion.
$[(p \to q) \land \neg p] \to \neg q$	Fallacy of denying the hypothesis.

- **Example**:

	$p \to q$	If it's raining, then the ground is wet
	q	The ground is wet.
$\therefore$	p	Therefore, it's raining. (Wrong: wet ground can be caused by many different reasons.)

This is an example of an incorrect/invalid argument caused by the fallacy of affirming the conclusion.

DO NOT do inference like this!!!

Examples of Inference 5

- **Fallacies**: Incorrect reasoning

Proposition	Description
$[(p \to q) \land q] \to p$	Fallacy of affirming the conclusion.
$[(p \to q) \land \neg p] \to \neg q$	Fallacy of denying the hypothesis.

- **Example**:

	$p \to q$	If it snows today, then university will be closed.
	$\neg p$	It didn't snow today.
$\therefore$	$\neg q$	Therefore, the university will not be closed. (Wrong: on July 4 there is no snow, but the university will still be closed.)

This is an example of an incorrect/invalid argument caused by the fallacy of denying the hypothesis.

DO NOT do inference like this!!!

Examples of Inference 6

- **Example**: Identify the error or errors in this argument that supposedly shows that if $\exists x P(x) \vee \exists x Q(x)$ is true then $\exists x(P(x) \wedge Q(x))$ is true.

Step	Proposition	Explanation
1	$\exists \boldsymbol{x} \boldsymbol{P}(\boldsymbol{x}) \vee \exists x \boldsymbol{Q}(\boldsymbol{x})$	Premise
2	$\exists \boldsymbol{x} \boldsymbol{P}(\boldsymbol{x})$	Simplification from (1) Simplification is not from disjunction
3	$\boldsymbol{P}(C)$	Existential instantiation from (2)
4	$\exists \boldsymbol{x} \boldsymbol{Q}(\boldsymbol{x})$	Simplification from (1): same as (2)
5	$\boldsymbol{Q}(C)$	Existential instantiation from (4) We cannot assume that the C that makes $\boldsymbol{P}$ true is the same C that make $\boldsymbol{Q}$ true.
6	$\boldsymbol{P}(C) \wedge \boldsymbol{Q}(C)$	Conjunction from (3) and (5)
7	$\exists \boldsymbol{x}(\boldsymbol{P}(\boldsymbol{x}) \wedge \boldsymbol{Q}(\boldsymbol{x}))$	Existential generalization

Indirect Proof

- Previous inference is the method of direct proof
- We will also learn two important indirect proofs
  - **Proof by contraposition**
    - Proof of the equivalent contrapositive
  - **Proof by contradiction**
    - Negation of the conclusion is a contradiction, which is **always false**, then the original conclusion is **always true**
    - First **assume the original conclusion is wrong**, and this is the "negation of the conclusion"
    - Then prove this assumption is a contradiction

Example of Proof by Contraposition 1

- Prove that if n is an integer and $n^3 + 5$ is odd, then n is even.

Step	Statements	Explanation
1	Let proposition $\boldsymbol{p}$ be "$\boldsymbol{n^3 + 5}$ is an odd integer."	Premise
2	Let proposition $\boldsymbol{q}$ be "$\boldsymbol{n}$ is an even integer."	Conclusion
3	To prove $\boldsymbol{p \rightarrow q}$, we can try to show the contrapositive $$\neg \boldsymbol{q} \rightarrow \neg \boldsymbol{p}$$ i.e., we can show that if "$\boldsymbol{n}$ is odd," then "$\boldsymbol{n^3 + 5}$ is even."	Definition of contrapositive.
4	$\neg \boldsymbol{q}$ means "$\boldsymbol{n}$ is odd", and let $\boldsymbol{n = 2k + 1}$, where $\boldsymbol{k}$ is an arbitrary integer.	Premise of contrapositive. Definition of odd integer.

Example of Proof by Contraposition 2

- Prove that if n is an integer and $n^3 + 5$ is odd, then n is even.

Step	Expression	Justification
5	$\boldsymbol{n^3 + 5 = (2k+1)^3 + 5}$ $\boldsymbol{= 8k^3 + 12k^2 + 6k + 6}$ $\boldsymbol{= 2(4k^3 + 6k^2 + 3k + 3)}$ $= 2\boldsymbol{u}$	Substitution based on (4) Let $\boldsymbol{u = 4k^3 + 6k^2 + 3k + 3}$
6	$\boldsymbol{n^3 + 5 = 2u}$is an even integer, which is $\neg\boldsymbol{p}$	Definition of even integer
7	We have shown the contrapositive, that $\neg\boldsymbol{q} \rightarrow \neg\boldsymbol{p}$.	Step 4-6
8	$\boldsymbol{p} \rightarrow \boldsymbol{q}$, which is "if $n^3 + 5$ is odd, then n is even"	Equivalent contrapositive

Example of Proof by Contradiction

- Prove that if n is an integer and $n^3 + 5$ is odd, then n is even.

Step	Statements	Explanation
1	Let $\neg \boldsymbol{q}$ be the proposition that $\boldsymbol{n}$ is odd.	Assume the conclusion is false
2	$n^3 + 5$ is odd	Premise
3	$\boldsymbol{n^3}$ is odd	The product of three odd integers is odd, based on (1)
4	$\boldsymbol{n^3 + 5}$ is even	Sum of two odd integers is even, based on (3)
5	$\neg \boldsymbol{q}$ is false, i.e. $\boldsymbol{n}$ is even.	Contradiction, the integer $\boldsymbol{n^3 + 5}$ cannot be even and odd simultaneously, based on (2) and (4)

Practices 1

- 1. What conclusion(s) can be drawn?

  - "If I eat spicy foods, then I have strange dreams."
    "I have strange dreams if there is thunder while I sleep."
    "I did not have strange dreams."

  - "I am either clever or lucky."
    "I am not lucky."
    "If I am lucky, then I will win the lottery."

Practices 2

- 1. What conclusion(s) can be drawn?
  - "If I eat spicy foods, then I have strange dreams."
    "I have strange dreams if there is thunder while I sleep."
    "I did not have strange dreams."
    - I didn't eat spicy foods and there's no thunder while I sleep.
  - "I am either clever or lucky."
    "I am not lucky."
    "If I am lucky, then I will win the lottery."
    - I am clever.

Practices 3

- 2. Determine whether the argument is correct or not, and explain why.
  - Every computer science major takes discrete structures. Natasha is taking discrete structures. Therefore, Natasha is a computer science major.
  - All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.
  - Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola everyday.

Practices 4

- 2. Determine whether the argument is correct or not, and explain why.
  - Every computer science major takes discrete structures. Natasha is taking discrete structures. Therefore, Natasha is a computer science major.
    - Incorrect. This is a fallacy of affirming the conclusion.
  - All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.
    - Incorrect. This is a fallacy of denying the hypothesis.
  - Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola everyday.
    - Correct. Universal instantiation and modus ponens are used.

Practices 5

- 3. How to show that
  "It is not raining or Yvette has her umbrella, "
  "Yvette does not have her umbrella or she does not get wet," and "It is raining or Yvette doesn't get wet"
  can imply "Yvette does not get wet"?

Practices 6

- 3. How to show that
  "It is not raining or Yvette has her umbrella, "
  "Yvette does not have her umbrella or she does not get wet," and "It is raining or Yvette doesn't get wet"
  can imply "Yvette does not get wet"?
  - Let be r be "it is raining", u be "Yvette has her umbrella", and w be "Yvette gets wet".
  - The premises are $\neg r \vee u$, $\neg u \vee \neg w$, $r \vee \neg w$
  - By resolution, we can combine $\neg r \vee u$ and $\neg u \vee \neg w$ to get $\neg r \vee \neg w$. Then further combine with $r \vee \neg w$, and thus we can get $\neg w$, which means she doesn't get wet.

Practices 7

- 4. Prove
  - For real numbers, if $x + y \geq 2$, then $x \geq 1$ or $y \geq 1$.
  - At least ten of any 64 days chosen must fall on the same day of the week.

Practices 8

- 4. Prove
  - For real numbers, if $x + y \geq 2$, then $x \geq 1$ or $y \geq 1$.
    - Proof by contraposition: if $x \geq 1$ or $y \geq 1$ can't hold, then $x + y < 2$.
    - "$x \geq 1$ or $y \geq 1$ can't hold" means $\neg(x \geq 1 \vee y \geq 1)$
    - By De Morgan's laws, this further means $x < 1 \wedge y < 1$, and thus $x + y < 2$.
  - At least ten of any 64 days chosen must fall on the same day of the week.
    - Proof by contradiction: assume we can't have such ten days, then we can have at most nine days on each day of the week, and thus we can't at most 7*9 days = 63 days, but we need to arrange 64 days – contradiction.

Extension

- 1. Conjunctive Normal Form (CNF)
- 2. First Order Logic
- 3. Resolution Refutation